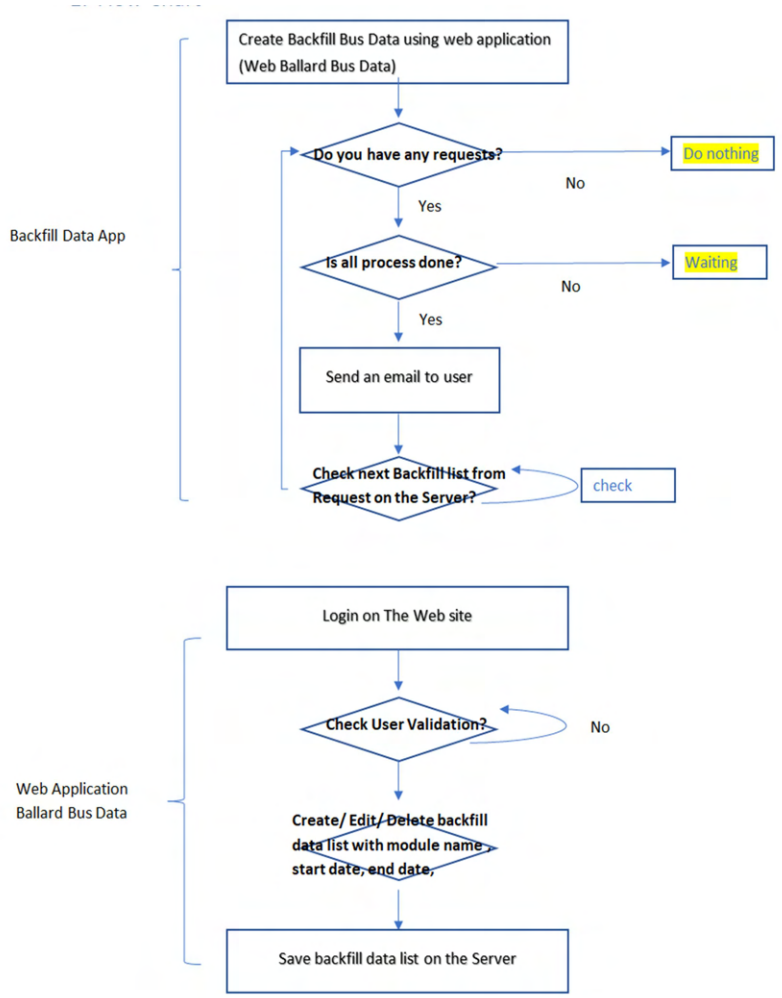
Portfolio

Andrew Kim

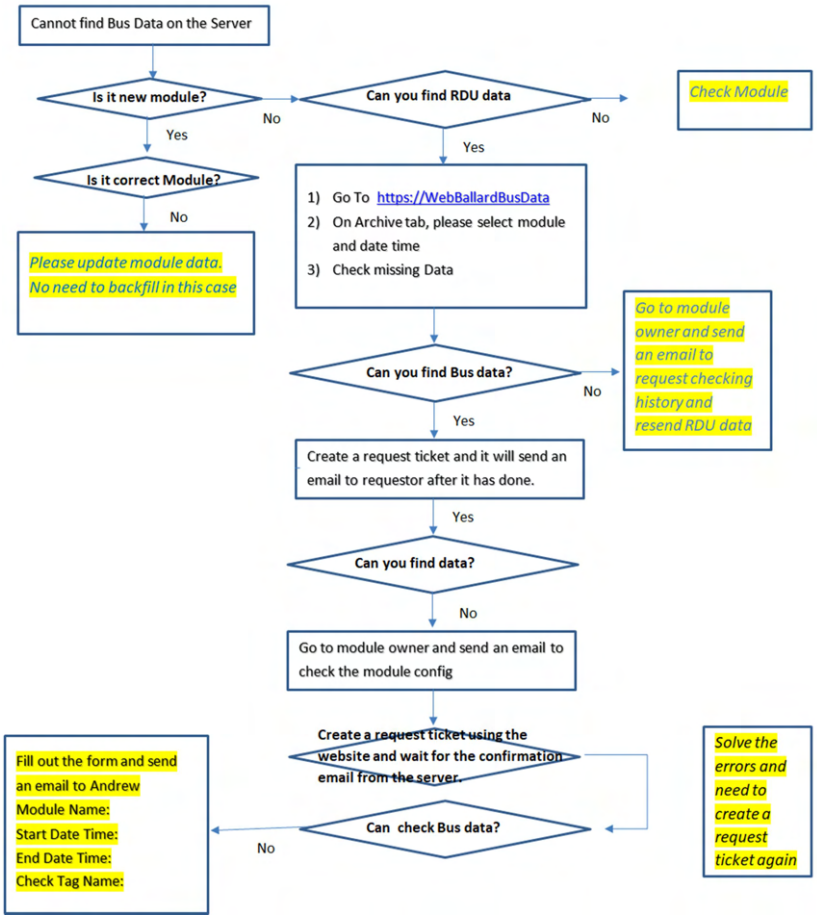
Ballard Bus Data

- As a project leader, I worked alongside executive leadership, project managers, control engineers, and end users to develop solutions to organizational needs.
- Developed a real-time data ingress system that download Bus data from FTP/SFTP server and convert DAT data into SQL data format. In addition, it imported approximately 50K files into SQL Local DB for Ballard, AVEVA Insight 3rd party server, and Azure Cloud Server every day.
- Maintained the Ballard Bus System in C# ASP.NET Core resulting in a remarkable 30 % reduction in processing time.
- Developed a web application to provide essential support for a controls team; therefore, it can check data loss, backfill missing data, and configure bus modules using a web application, resulting in a significant increase in overall team productivity.
- Technologies: C# ASP.NET MVC, VB, MS SQL, Ajax, Json, Git, Aveva, IIS and Azure.

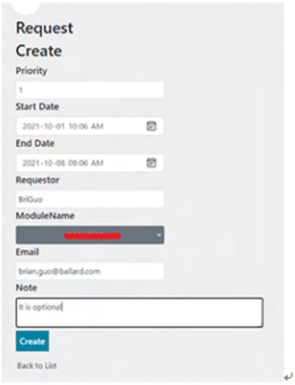
- **How to support Engineers and monitor the real-time data ingress system, Ballard Bus Data**



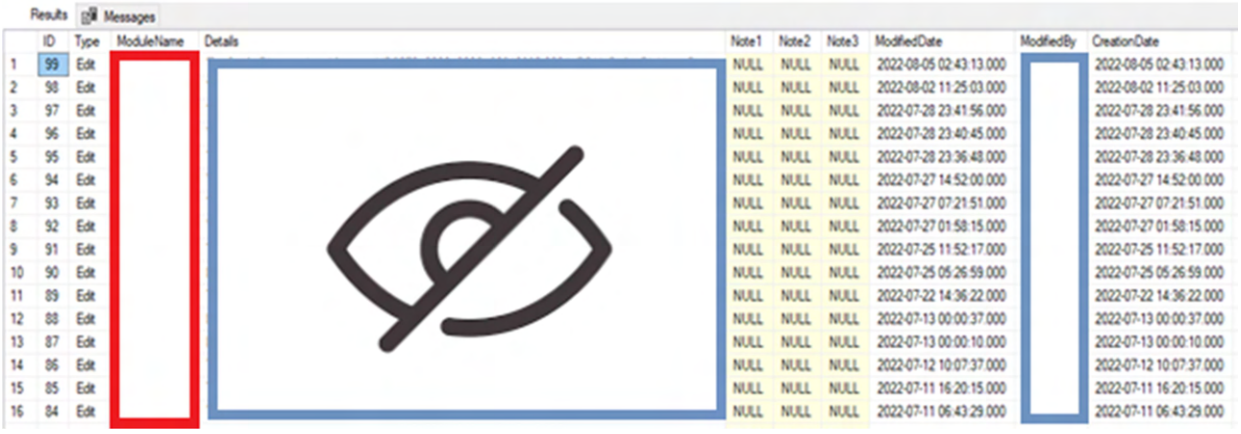
- **Troubleshooting using the WebBallardBusData**



- **Request Page:**
  Description: Create Backfilling data ticket by login user.
  Instruction: Once users create request tickets on the request page, it backfill data automatically and send an email with backfill results after complete.



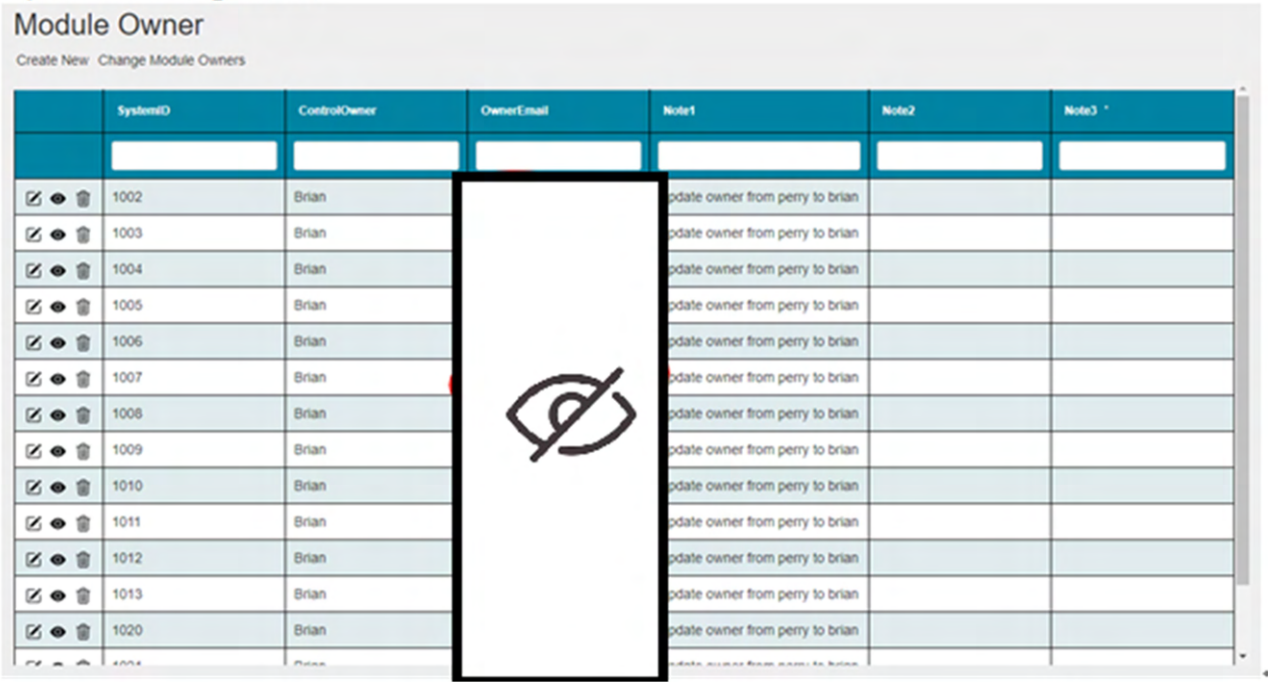
- **Module Page:**
  Description: Create/ Edit/ Delete module information
  Details: Configure all Ballard Bus Modules on the module page.



Results | Messages

	ID	Type	ModuleName	Details	Note1	Note2	Note3	ModifiedDate	ModifiedBy	CreationDate
1	99	Edit			NULL	NULL	NULL	2022-08-05 02:43:13.000		2022-08-05 02:43:13.000
2	98	Edit			NULL	NULL	NULL	2022-08-02 11:25:03.000		2022-08-02 11:25:03.000
3	97	Edit			NULL	NULL	NULL	2022-07-28 23:41:56.000		2022-07-28 23:41:56.000
4	96	Edit			NULL	NULL	NULL	2022-07-28 23:40:45.000		2022-07-28 23:40:45.000
5	95	Edit			NULL	NULL	NULL	2022-07-28 23:36:48.000		2022-07-28 23:36:48.000
6	94	Edit			NULL	NULL	NULL	2022-07-27 14:52:00.000		2022-07-27 14:52:00.000
7	93	Edit			NULL	NULL	NULL	2022-07-27 07:21:51.000		2022-07-27 07:21:51.000
8	92	Edit			NULL	NULL	NULL	2022-07-27 01:58:15.000		2022-07-27 01:58:15.000
9	91	Edit			NULL	NULL	NULL	2022-07-25 11:52:17.000		2022-07-25 11:52:17.000
10	90	Edit			NULL	NULL	NULL	2022-07-25 05:26:59.000		2022-07-25 05:26:59.000
11	89	Edit			NULL	NULL	NULL	2022-07-22 14:36:22.000		2022-07-22 14:36:22.000
12	88	Edit			NULL	NULL	NULL	2022-07-13 00:00:37.000		2022-07-13 00:00:37.000
13	87	Edit			NULL	NULL	NULL	2022-07-13 00:00:10.000		2022-07-13 00:00:10.000
14	86	Edit			NULL	NULL	NULL	2022-07-12 10:07:37.000		2022-07-12 10:07:37.000
15	85	Edit			NULL	NULL	NULL	2022-07-11 16:20:15.000		2022-07-11 16:20:15.000
16	84	Edit			NULL	NULL	NULL	2022-07-11 06:43:29.000		2022-07-11 06:43:29.000

- **Module Owner Page:**
  Description: Check module owners in Controls team and easy to communicate with them
  Details:
  Filtering/ sorting module owner table. Create new owner and update existing owner.

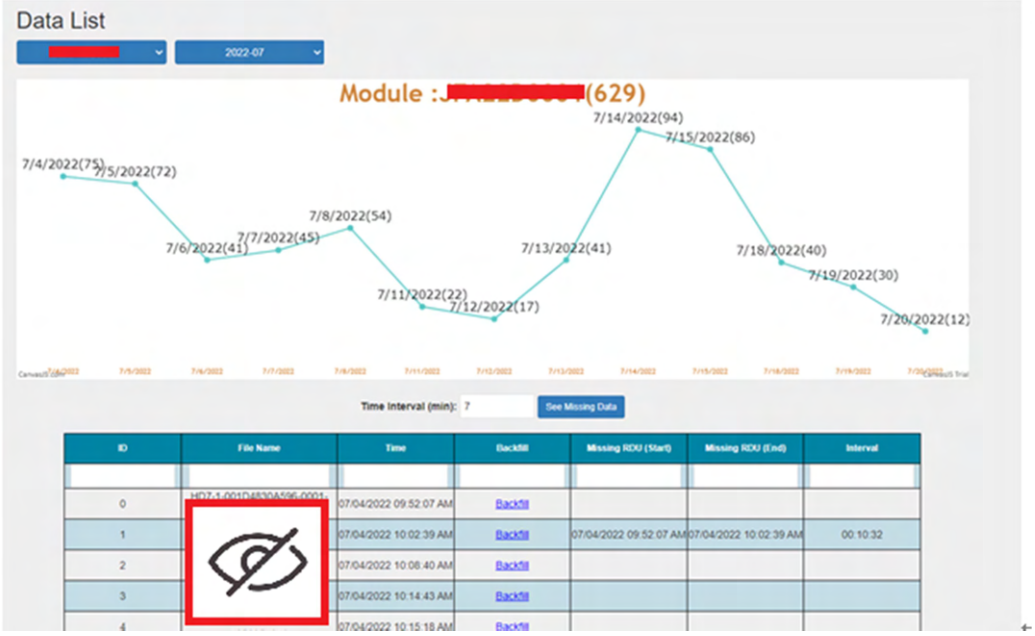


Module Owner

Create New | Change Module Owners

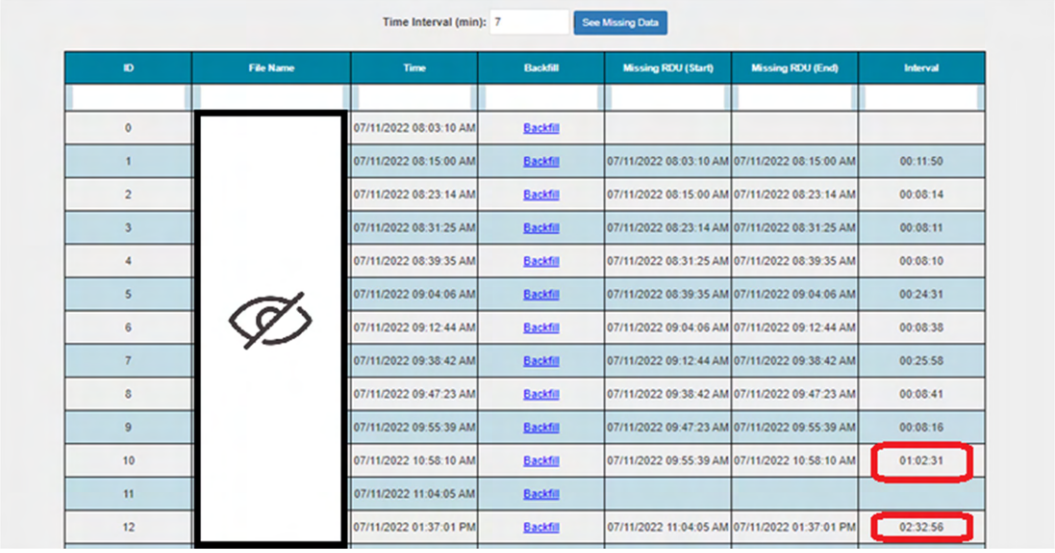
	SystemID	ControlOwner	OwnerEmail	Note1	Note2	Note3
	1002	Brian		pdate owner from perry to brian		
	1003	Brian		pdate owner from perry to brian		
	1004	Brian		pdate owner from perry to brian		
	1005	Brian		pdate owner from perry to brian		
	1006	Brian		pdate owner from perry to brian		
	1007	Brian		pdate owner from perry to brian		
	1008	Brian		pdate owner from perry to brian		
	1009	Brian		pdate owner from perry to brian		
	1010	Brian		pdate owner from perry to brian		
	1011	Brian		pdate owner from perry to brian		
	1012	Brian		pdate owner from perry to brian		
	1013	Brian		pdate owner from perry to brian		
	1020	Brian		pdate owner from perry to brian		

- **Description: Check Bus data and display them with a line chart. Easy to find missed data and backfill RDU data using the web application**
  Instruction: Select a module and year-month ~ Enter time interval, default setting is 7 min~ Click 'See Missing Data' to check data
  Result: it did not send JFA22D0001 data between 07/04/2022 9:52 and 07/04/2022 10:02 but, it is acceptable and ignore

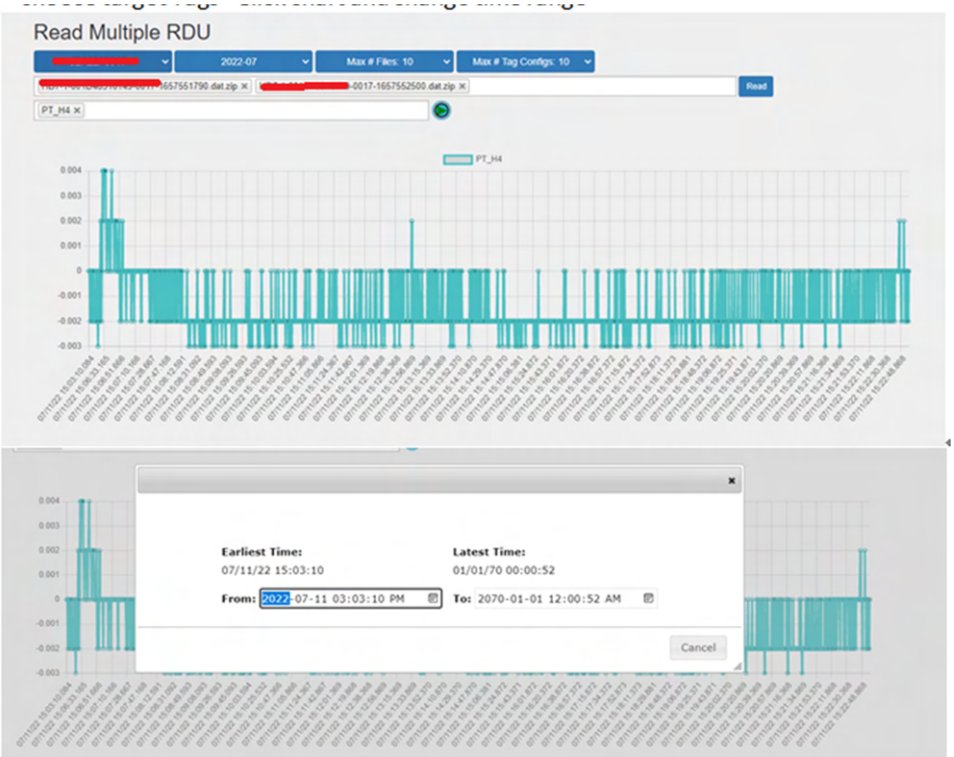


it.

Result: it did not send data between 07/11/2022 11:04 AM and 07/11/2022 01:37 PM. for 2.5 hours and it needs to investigate. Warning missing Dada and check module JBA22F0017 log files with Controls team.



- **Read Ballard Bus Data Page:**
  Description: Read RDU data with line charts
  Instruction: Select 'Module' ~ Select 'year-month' ~ choose RDU files from archive folder ~ choose target Tags~ Click chart and change time range



- **Status Page:**
  Description: Monitor all the instances. / update the page every 10 secs

  - **Received and waiting: RDU files in processing**
  - **In processing Folder: Upload data to Local DB**
  - **Not processed: May have an error/ delay to upload Historian**
  - **Waiting for Move/ Waiting for upload: Upload data to Azure cloud Server**

- **History Records Page:**
  Description: Show log files from BPSQ62 and cannot be editable.
  - HistoryLog – Check NonFileNameFilter folder and move valid RDU data to Chameleon folder

Select a Table

HistoryLog | ALL

ErrorType1	ErrorType2	ModuleName	Details	Result	IsSolved	Note1	Note2	Note3	ModifiedDate	ModifiedB
		ALL							From yyyy-mm-dd To yyyy-mm-dd	
Maintain	Check and move file from NonConfigFiles to Chameleon	ALL	(11/63020) files are moved.		0				2022-08-05 3:02:01 PM	
Maintain	Check and move file from NonConfigFiles to Chameleon	ALL	(6/63891) files are moved.		0				2022-08-05 10:56:42 AM	
Maintain	Check and move file from NonConfigFiles	ALL	(0/63872) files are moved.		0				2022-08-05 9:33:03 AM	

  - HistoryModule – Track all history when users update module data

Select a Table

HistoryModule

	Type	ModuleName	Details	Note1	Note2	Note3	ModifiedDate	ModifiedBy	CreationDate
		KAA21J0002					From yyyy-mm-dd To yyyy-mm-dd		From yyyy-mm-dd To yyyy-mm-dd
58	Edit	[redacted]					2022-05-20 1:36:44 AM	jusbui	2022-05-20 1:36:44 AM
59	Edit	[redacted]					2022-05-20 1:38:32 AM	jusbui	2022-05-20 1:38:32 AM

  - ErrorLog – Related to error when uploading data to Historian and Wonderware
    Need to add missing tag configuration file and re-upload module HBA22C0010 data.

Select a Table

ErrorLog | 3014

	SystemID	Error	TagConfigFile	Details	IsSolved	Note1	Note2	Note3	Modi
	HBA22C0010								From yyyy-r To yyyy-r
225	[redacted]	[redacted]ConfigFile		Not existed	0				2022-0 8:23:3

  - Request – Check request ticket and result

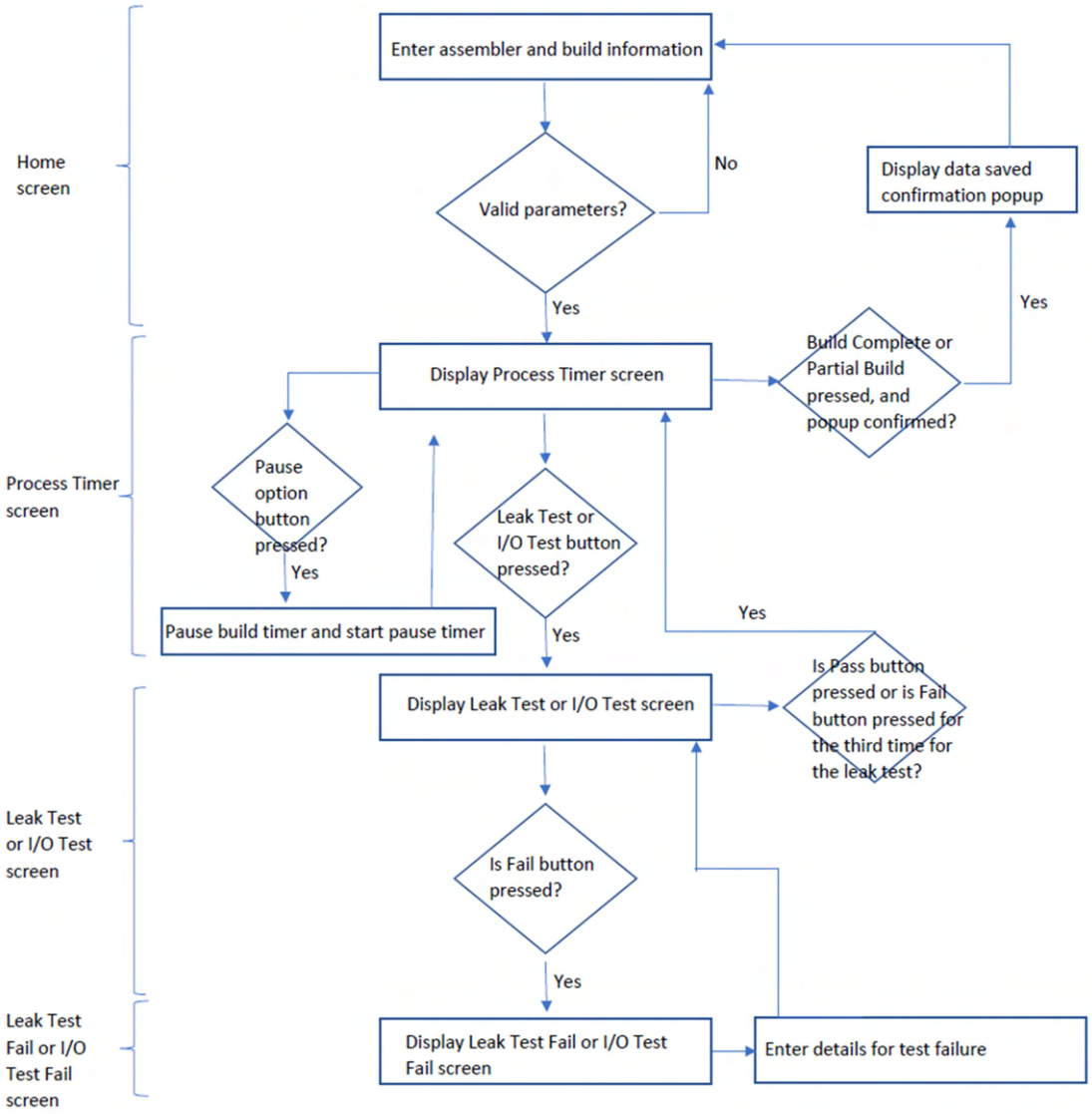
Request

	Priority	StartDate	EndDate	Requestor	ModuleName	Email	CreationDate	Result	Note	Active
		From yyyy-mm-dd To yyyy-mm-dd	From yyyy-mm-dd To yyyy-mm-dd		DB204222C000		From yyyy-mm-dd To yyyy-mm-dd			
24167	1	2022-07-27 9:36:00 AM	2022-07-28 9:36:00 AM				2022-07-28 9:36:56 AM	COMPLETE	Completed 232 files. - Jul 28 2022 9:43AM	0

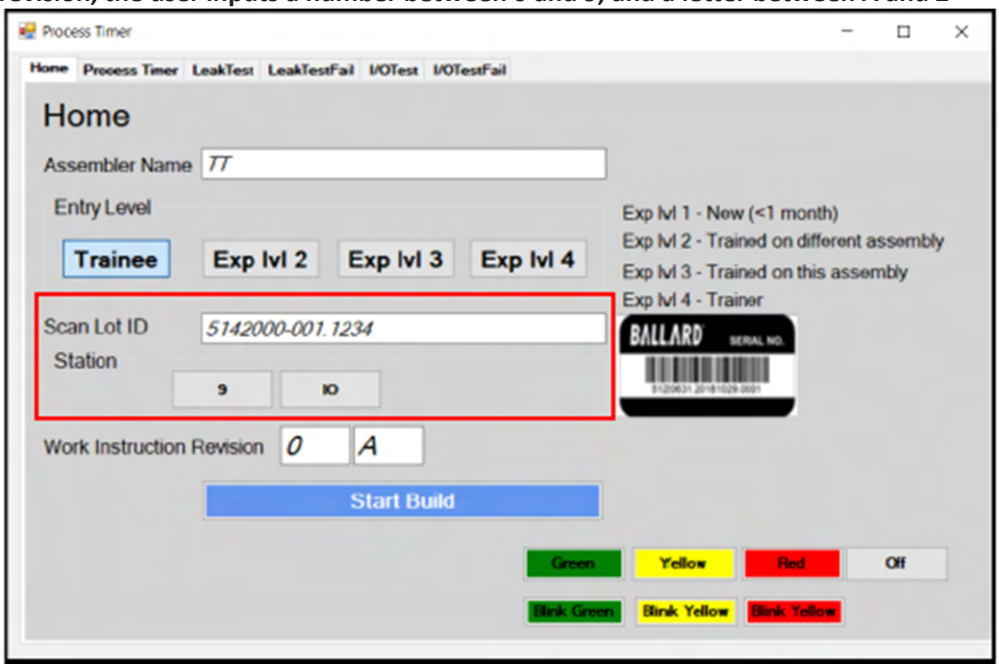
Module Assembly Timer

- As a project leader, I made a project schedule and managed a project development meeting.
- Designed and developed DB schema and applications using C# .NET MVC.
- Collected all task time and analyzed the working patterns. It helped understanding what their tasks are and issues.
- Designed entity-relationship model and created test cases and test plan and performed manual testing of a web-based application.
- Technologies: ASP.NET Core, C#, MS SQL, T-SQL, Ajax, JSON, JavaScript, SVN, and IIS
- **The main purpose of Stack Timer is to track the time for module assembly. With this application, the assembler can record the time it took to do a complete or partial build and the data can later be analyzed to find improvements and monitor issues. The application handles special cases depending on the Lot ID, provides functionality for pausing and resuming the build timer, and allows the assembler to input the results of leak and I/O tests. Also, the application changes the stack light colour based on the remaining time for a build and which screen is being displayed. The application is a Windows Form written in C# with Entity Framework 6. The database is ModuleAssembly and it can be found in the local server using SSMS.**

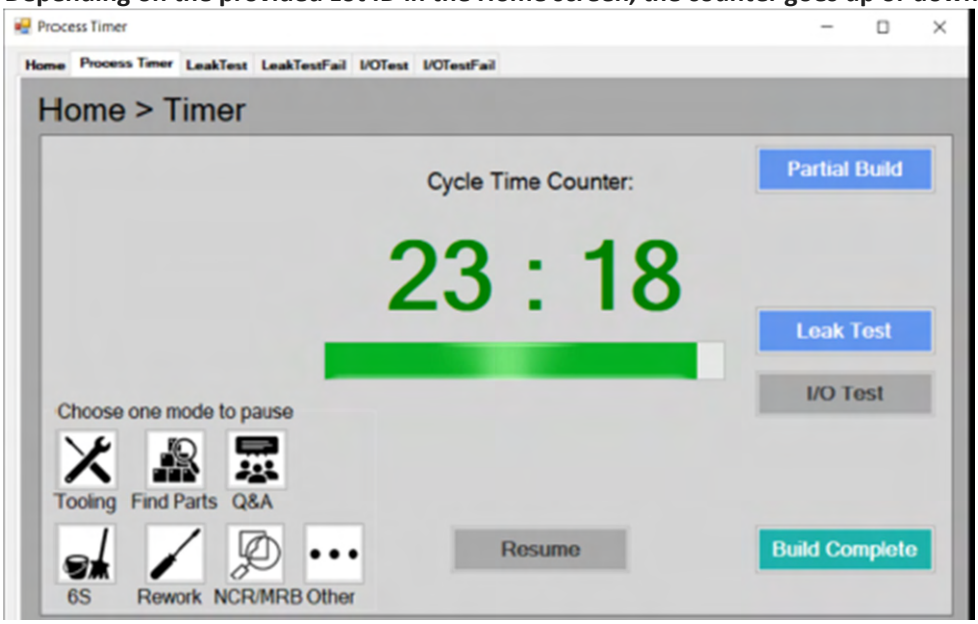
1. Flow Chart



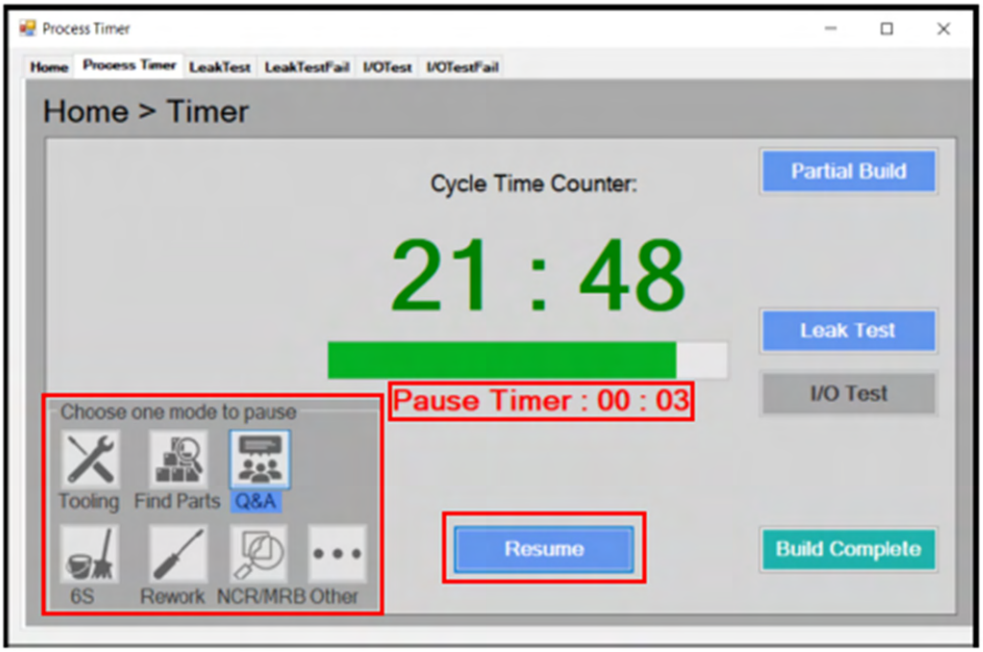
- **Home Screen:**
  Description: Enter required information to start or continue a build
  Details: The assembler enters their initials, experience level, the Lot ID, and revision For the revision, the user inputs a number between 0 and 9, and a letter between A and Z



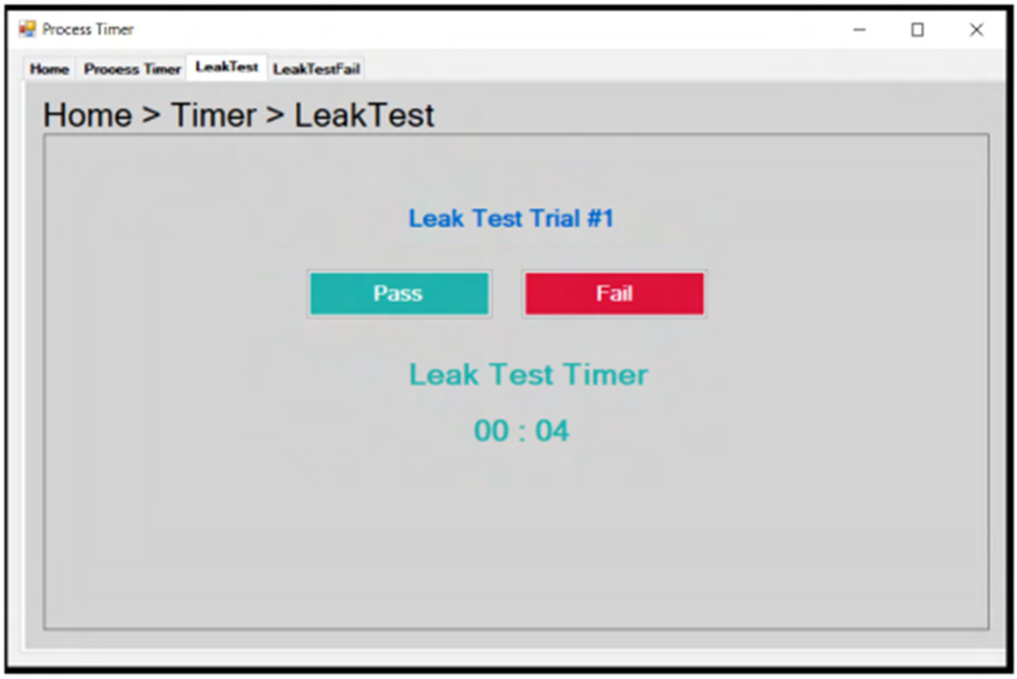
- **Process Timer Screen:**
  Description: Displays the build time, pause or end build
  Details: The Assembler can see a counter with the time for a build and a progress bar Depending on the provided Lot ID in the Home screen, the counter goes up or down



Pause Option:
The user can pause the counter by choosing one of the pause options and, when the counter is paused, a different counter is displayed to track the paused time of the build To resume the build, the Resume button must be pressed

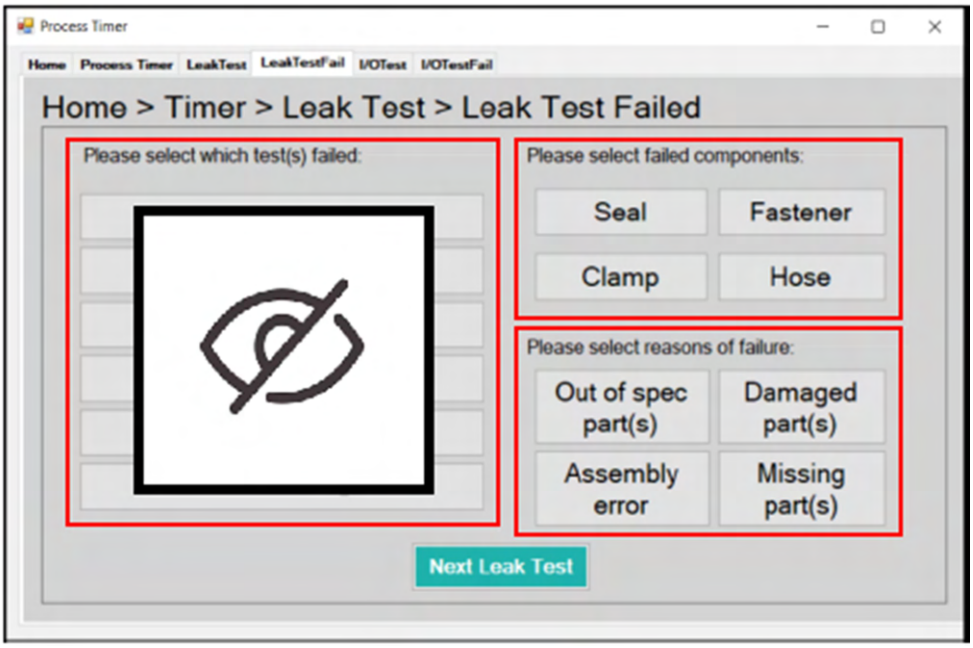


Leak Test Screen:
Description: Input the results of a leak test trial
Details: The assembler can input the results of a leak test trial by pressing the Pass or the Fail buttons If the Pass button is pressed, then the application directs the user to the Process Timer screen If the Fail button is pressed, then the application directs the user to the Leak Test Fail screen The leak test has its own timer, but the data is further processed to separate the time taken for each trial



Leak Test Fail Screen:
Description: Input the details of a failed leak test trial
Details: The content of this screen is dependent on the Lot ID provided in the Home screen The

assembler can specify which tests failed, the failed components, and the reasons of failure



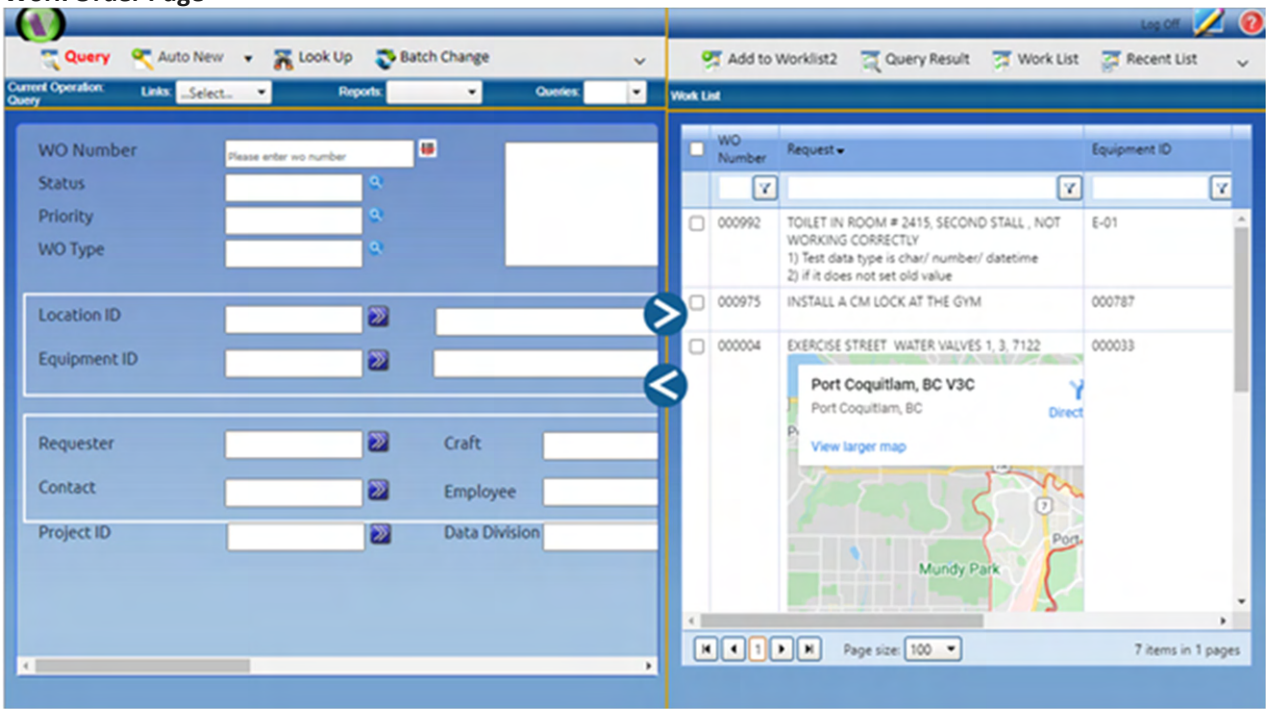
- **Stack Light Options:**
  - **Off, Green, Yellow, Red, Blink Green, Blink Red**

Azzier CMMS

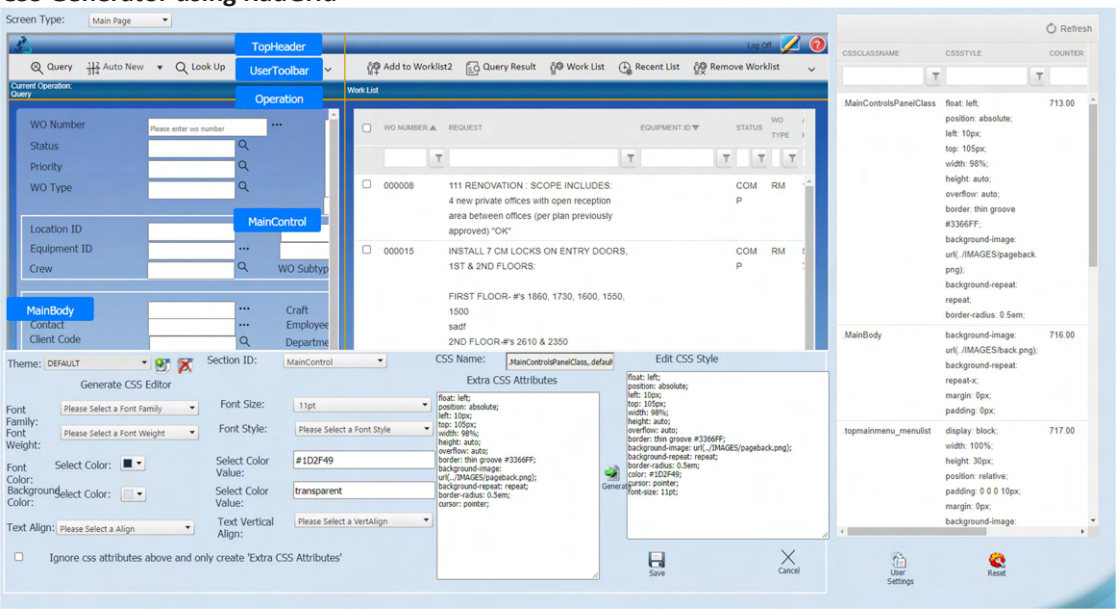
- Worked collaboratively with a customer service team to develop and maintain the CMMS software.
- Provided better manage, predict, and optimize clients' assets.
- Helped to boost operating efficiency with maintenance management.
- Technologies: ASP.NET, C#, MS SQL Server 2014, Ajax, JSON, JavaScript, SVN, WEB API, and CORS

- **Azzier CMMS Software**
  My role was to develop, test, and maintain the CMMS software to provide better manage, predict, and optimize clients' assets, operations. There are two significant projects for me. The first one is to develop user interface for css style; therefore, user can decorate the web application using css editor. The most challenging part was to customize Telerik UI for RadGrid and KendoGrid because it was difficult to automatically sort CSS properties from client-side. In addition, I need to consider performance and efficiency without creating external css style; therefore, I used getComputedStyle method to get computed value of style property and populate them on the screen. In addition, every user can save their style into user settings. The second project is to develop Android application using Cross-Origin Resource sharing so that it can transfer date from master domain to client domain. Login/ Validation company key code/ download and upload data/ uploading files and update data with specific process.

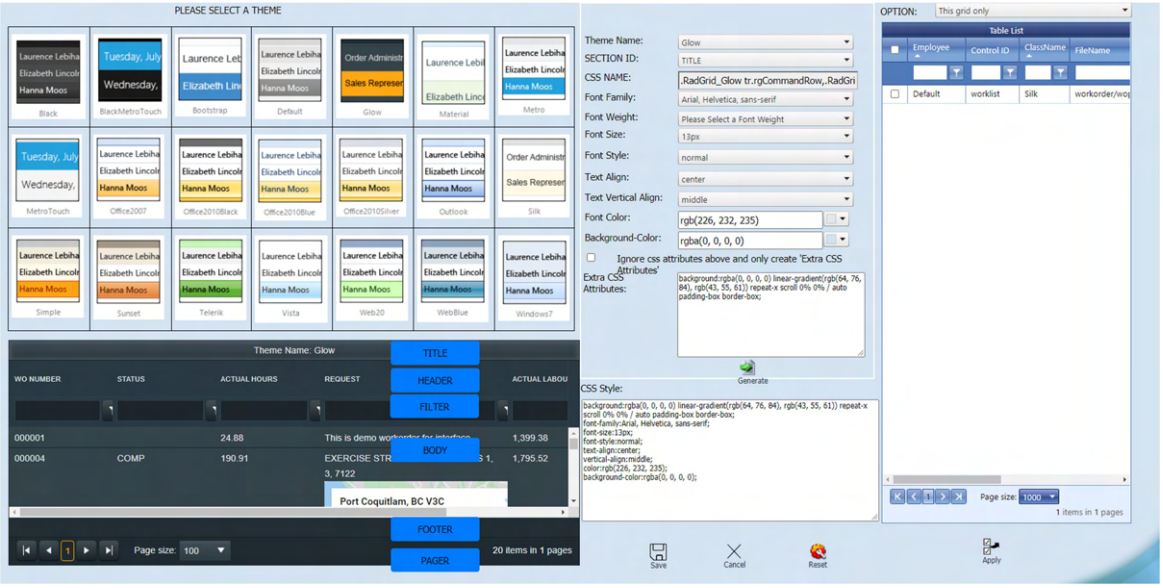
Work Order Page



CSS Generator using RadGrid

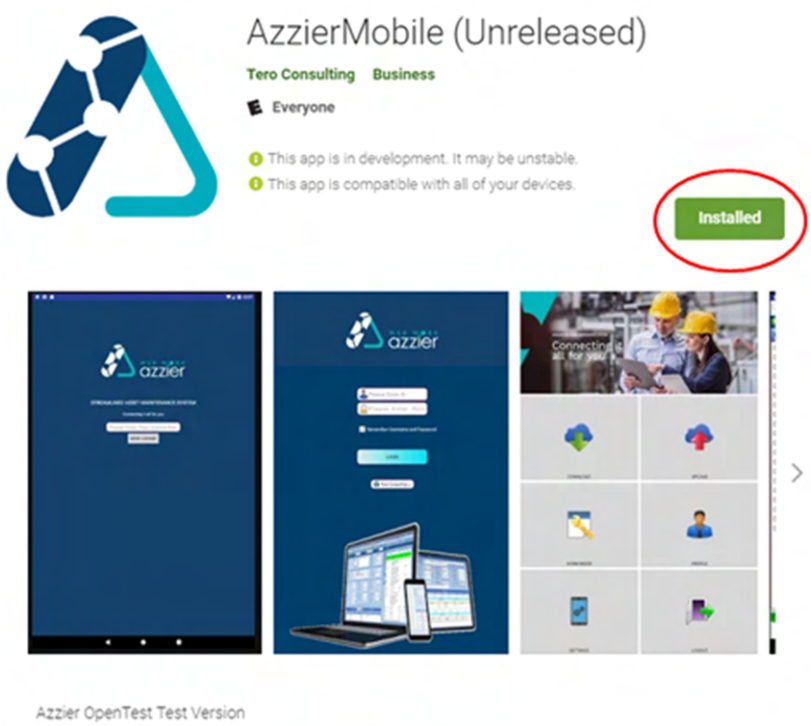


CSS Theme with Editor

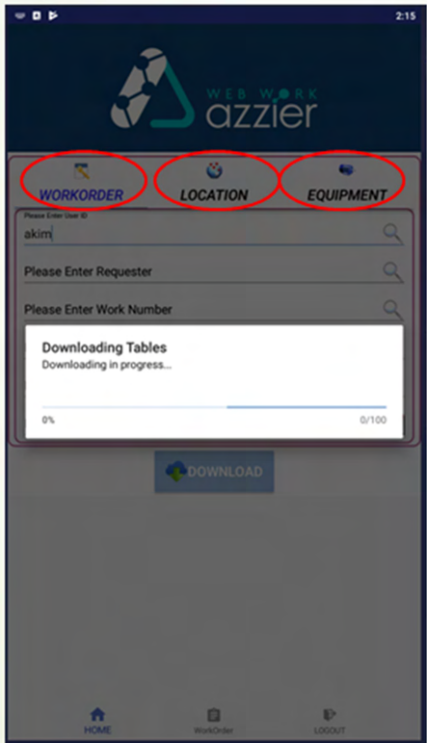


- **Azzier Android Application**

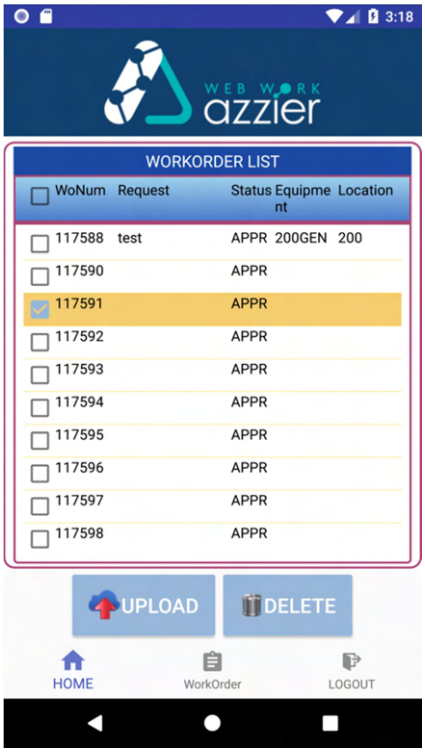
Azzier Mobild on Google Play



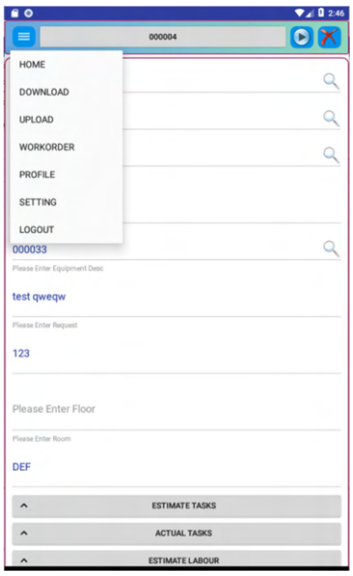
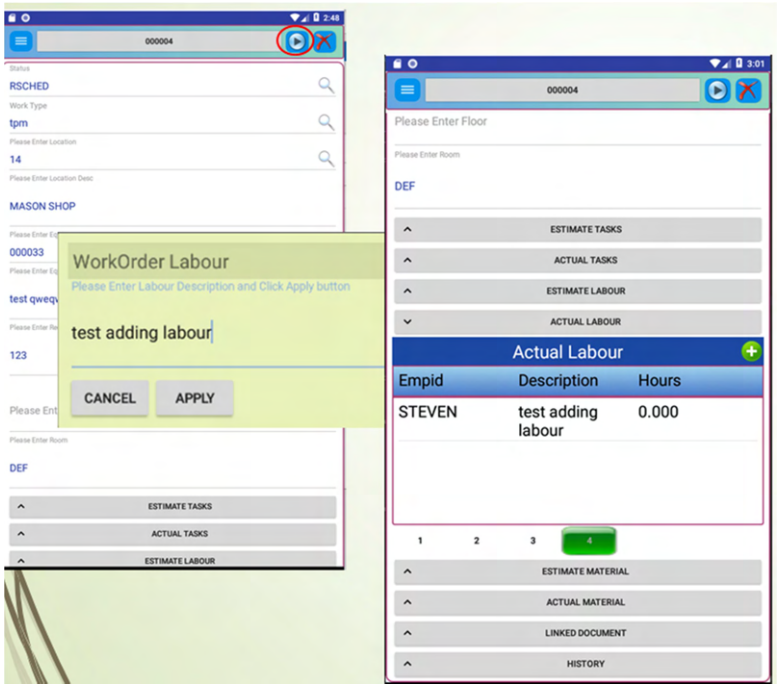
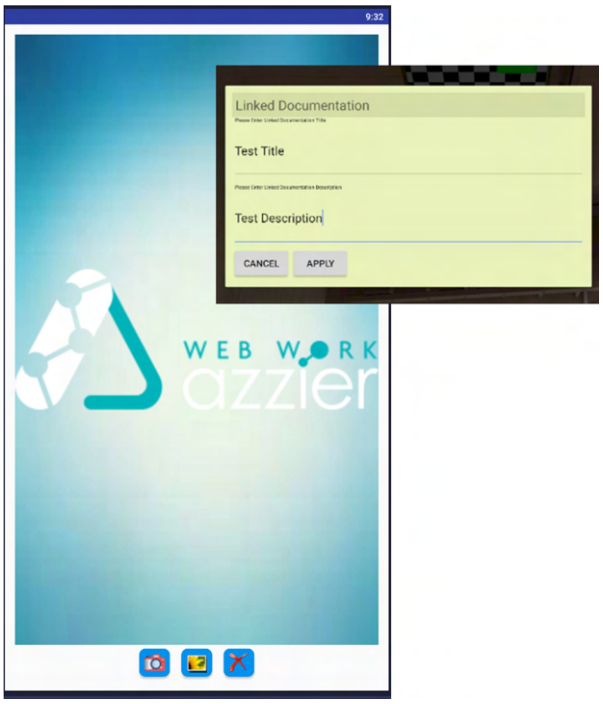
Download Work Order



Check Work Order



Work Order



- Edit Status/ Work Type/ Location/ Location Desc/ Equipment/ Equipment Desc/ Request/ Note2/ Room From Workorder
- Click 'Start' button to generate WorkOrder Labour
- Add Actual Tasks/ Actual Labour/ Actual Material/ Linked Documentation
- Check only downloaded resources like Tasks/ Labour/ Material/ WoHistory (not editable)
- Create new Linked Docs with picture, title, and description using camera